

Analysis PhD Exam, May 2005

Give complete proofs and computations. Be particularly careful to indicate why the most crucial steps in your proof are true.

1. Let μ be a positive Radon measure on a locally compact Hausdorff space T . Let $f_n : T \rightarrow \overline{\mathbb{R}}$ be μ -measurable for all $n \in \mathbb{N}$. Prove that $\sup\{f_n \mid n \in \mathbb{N}\}$ is μ -measurable.
2. Let E be a complex Banach space. Suppose $\{x_n\}_{n \in \mathbb{N}} \subset E$ converges in the weakened topology $\sigma(E, E')$ to $x \in E$, i.e. $|\langle x_n - x, y \rangle| \rightarrow 0$ for all $y \in E'$ as $n \rightarrow \infty$. Prove that x is contained in the closure of the linear span of the set $\{x_n\}_{n \in \mathbb{N}}$.
3. Let E be a topological vector space. Prove the following:
 - (a) If $A \subset E$ is arbitrary and $U \subset E$ is open, then $A + U$ is open.
 - (b) If $A \subset E$ is compact and $B \subset E$ is closed, then $A + B$ is closed.
 - (c) The sum of closed subsets may fail to be closed.
4. Let $\delta \in \mathcal{D}'(\mathbb{R})$ denote the distribution given by $\delta(\phi) = \phi(0)$, for all $\phi \in \mathcal{D}(\mathbb{R})$. Prove your answers are correct.
 - (a) For which $f \in C^\infty(\mathbb{R})$ is $f \cdot \delta' = 0$?
 - (b) What is the support of δ ?
 - (c) If $X \in \mathcal{D}'(\mathbb{R})$, $f \in C^\infty(\mathbb{R})$, and f restricted to the support of X is 0, is it true that $f \cdot X = 0$?
5. Let $K : [0, 1] \times [0, 1] \rightarrow \mathbb{R}$ be continuous and $f : [0, 1] \times \mathbb{R} \rightarrow \mathbb{R}$ be continuous and bounded. Show that the Hammerstein equation

$$u(s) = \int_0^1 K(s, t) f(t, u(t)) dt, \quad s \in [0, 1],$$

has a solution in $E = \{x : [0, 1] \rightarrow \mathbb{R} \mid x \text{ continuous}\}$.

6. Let $1 \in \mathcal{D}'(\mathbb{R})$ be given by $1(\phi) = \int_{\mathbb{R}} \phi dx$, for all $\phi \in \mathcal{D}(\mathbb{R})$. Compute $(1 * \delta')(\phi)$ for all $\phi \in \mathcal{D}(\mathbb{R})$.
7. Let μ be a σ -finite positive Radon measure on a locally compact Hausdorff space T . Let $\{f_n\}_{n \in \mathbb{N}} \subset L^p(T, \mu)$ for some $p \in [1, \infty)$. Suppose that $\lim_{n \rightarrow \infty} f_n(t) = f(t)$ μ -a.e., that $f \in L^p(T, \mu)$, and that $\lim_{n \rightarrow \infty} \int |f_n|^p d\mu = \int |f|^p d\mu$. Prove that $\lim_{n \rightarrow \infty} \int |f_n - f|^p d\mu = 0$. Hint: Bring the Lebesgue Dominated Convergence Theorem and the σ -finiteness of the measure into play.
8. Let $\mathcal{H}$ be a Hilbert space with inner product $(\cdot, \cdot)$, and let $\{y_\alpha\}_{\alpha \in A} \subset \mathcal{H}$ be an orthonormal family. Let $x \in \mathcal{H}$.
 - (a) Prove that $\sum_\alpha (x, y_\alpha) y_\alpha$ converges strongly to an element $y \in \mathcal{H}$.
 - (b) Prove that the vector $x - y$ is orthogonal to the closure of the linear span of the set $\{y_\alpha\}_{\alpha \in A}$.